

Sreelekha Guda

Software Engineer – Java, Spring Boot, Distributed Systems & Microservices – Apache Kafka – AWS
9381472895 | sreelekha2.guda@gmail.com | Claude Certified Architect | LinkedIn | GitHub | LeetCode 1800+

Summary

Software Engineer with 1.5+ years of professional experience and deep, self-directed depth in **Java and Spring Boot distributed systems**. Designed and built **Vykronis**, a 9-service Spring Boot / Apache Kafka autonomous incident-response platform (14K+ LOC, 298 tests, fail-closed policy engine, human-approval-gated remediation), demonstrating production-grade architecture, event-driven design, and testing discipline. Professionally, built scalable backend microservices and REST APIs (Python/FastAPI, Django), delivered third-party integrations (ZoomInfo, Zoho), and shipped production AI/LLM retrieval systems. Comfortable across relational/NoSQL databases, Docker, CI/CD, AWS (EC2, S3, VPC, Lambda), and React.js full-stack development.

Technical Skills

Languages: Java, Python, JavaScript, TypeScript, SQL

Backend / Distributed Systems: Java Spring Boot (SpringBoot), Spring Cloud, Spring Data JPA / Hibernate, Spring Security, Spring AI, FastAPI, Django, Flask, REST APIs, Microservices, Event-Driven Architecture

Messaging & Streaming: Apache Kafka, Kafka Streams, KRaft, RabbitMQ, Event Buses, Topic/Schema Design

Databases: PostgreSQL (Flyway migrations), MySQL, MongoDB, Redis, Elasticsearch

Cloud & DevOps: AWS (EC2, S3, VPC, Lambda, Serverless, EBS), Docker, Docker Compose, Helm / kind, GitHub Actions, GitLab CI/CD, GraalVM Native Image

Frontend: React.js, Next.js, TypeScript, Responsive UI

Testing & Observability: JUnit, Mockito, Pytest, OpenTelemetry, Unit & Integration Testing (46%+ test-to-code ratio on flagship project)

AI/ML Engineering: RAG Pipelines, Embeddings, Prompt Engineering, Tool-Constrained AI Agents, LLM Evaluation (LangFuse, DeepEval), Claude (Certified Architect)

Core Concepts: Data Structures & Algorithms, OOP & Design Patterns, System Design, Distributed State Machines, Policy-Based Access Control

Projects

Vykronis – Autonomous Incident-Response Platform ↗ *Java 25, Spring Boot 4, Apache Kafka, PostgreSQL, Redis, Spring AI, Docker, Kubernetes*

- Architected and built a closed-loop, self-healing operations platform of **9 Spring Boot microservices** (14 Maven modules, 13-container Docker Compose stack) that detects, investigates, and remediates production incidents with a human-approval gate – **14,324 lines of Java** across 247 files.
- Designed an event-driven core over **Apache Kafka** with **6 typed topics** and versioned JSON-Schema contracts spanning metrics, traces, alerts, deployments, and remediation commands; built a **Kafka Streams correlation engine** (60s windowed detection) converting raw telemetry into ranked, severity-classified incidents.
- Built a **9-state incident state machine**, persisted in **PostgreSQL with Flyway migrations**, with every state transition fully auditable, and a **fail-closed policy decision matrix** requiring a named human approver before any production remediation executes.
- Engineered a **tool-constrained AI investigation agent** (Spring AI) with deterministic rule-based fallback, so root-cause hypotheses are never fabricated and the pipeline never hard-fails when infrastructure is degraded.
- Wrote **298 unit and integration tests across 70 test classes (46% test-to-code ratio)**, covering state-machine transitions, Kafka topologies, and policy decisions.
- Verified the full closed loop end-to-end: injected a synthetic error burst and drove OPEN → HYPOTHESIS_READY → AWAITING_APPROVAL → REMEDIATING in **under 1 minute** on first try, on a self-hosted, zero-cost stack (Ollama, Kafka, Postgres, Redis) hardened to run on a **7.9 GB laptop**.
- Built **4 GitHub Actions CI/CD workflows**, including a live **kind-cluster smoke test** validating the Helm-deployed stack on every commit, plus a GraalVM native-image build.

URL Shortener Platform *Java, Spring Boot, MongoDB, AWS (EC2)*

- Built a Java Spring Boot backend microservice (Spring MVC, Spring Data JPA/Hibernate, Spring Security) handling **100+ requests/sec** and **5,000+ shortened links**; tested with **JUnit/Mockito** and deployed on **AWS EC2** (Free Tier).

Appointments Made Easy *Node.js, Express.js, ReactJS, MongoDB, HTML5/CSS3*

- Built a full-stack appointment management platform with a **React.js** front end (hand-written responsive UI in HTML5/CSS3) and Node.js/Express.js backend, load-tested to support **1,000+ concurrent users**.

AI-Powered Funnel CRM *FastAPI, RabbitMQ, PostgreSQL, React, RAG, LLM*

- Built an AI-assisted CRM automation platform qualifying **30,000+ leads/month** through event-driven (RabbitMQ) async pipelines, cutting infrastructure costs **85%**; implemented RAG-based lead classification with LangFuse/DeepEval quality monitoring.

Experience

PalTech Consulting Private Limited

Dec 2024 – Present

- Cut processing latency **80% (15s → 3s)** by re-architecting sequential data pipelines into concurrent, event-driven async workflows using FastAPI, Django, PostgreSQL, MongoDB, Redis, and RabbitMQ.
- Reduced duplicate task execution and workflow failures by **50%** through fault-tolerant processing, checkpoint persistence, and retry logic.
- Built scalable backend microservices and REST APIs processing **30,000+ records/requests per month** across PostgreSQL, MongoDB, and Redis.
- Delivered production integrations with **2 external data providers, including ZoomInfo**, handling schema validation, transformation, and error handling; safeguarded CRM data integrity, catching **100%** of invalid records before syncing to Zoho.
- Shipped production RAG-based LLM retrieval systems (embeddings, vector search, prompt engineering), improving retrieval relevance from **70% to 90%** via LangFuse/DeepEval evaluation.
- Automated deployments and database migrations via Docker, GitLab CI/CD, and **AWS (EC2, EBS, S3, Lambda)**, cutting deployment time **80%**. Collaborated in an **Agile/Scrum** team through sprints and code review.

Education

CMR College of Engineering & Technology	2021 – 2025
B.Tech, Computer Science Engineering (AI & ML)	CGPA: 9.24/10

Achievements

- 1800+ LeetCode rating; solved 700+ Data Structures & Algorithms problems.
- Claude Certified Architect • Finalist, IIITH Megathon & BVRIT Hackathon • Top performer, JNTUH Blind Coding Competition.